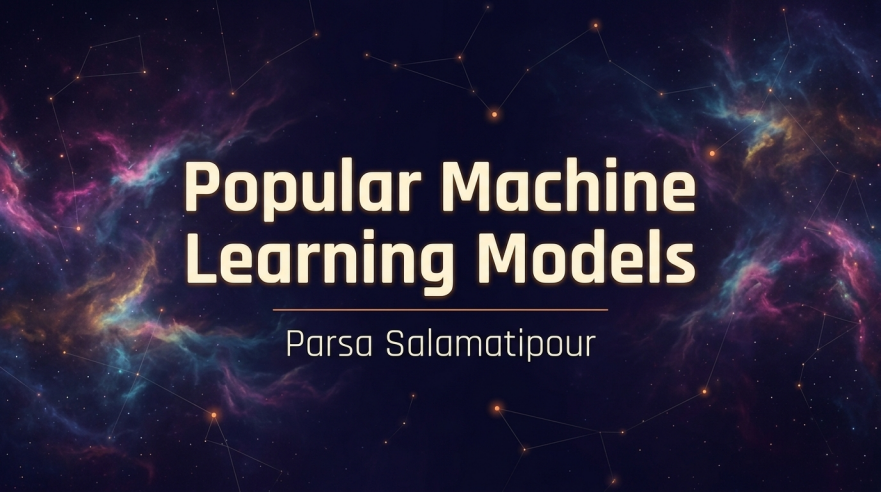


The background of the slide is a deep space image featuring vibrant nebulae in shades of blue, purple, and orange. Overlaid on this are several constellations, represented by thin white lines connecting small, bright yellow and orange stars. The overall effect is a celestial and mysterious atmosphere.

Popular Machine Learning Models

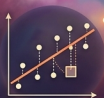
Parsa Salamatipour

Our Journey Today

How a Model Learns



How a Model Learns



Linear Regression

Decision Tree



Decision Tree



Loss Functions

K-Nearest Neighbours

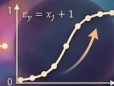


K-Nearest Neighbours

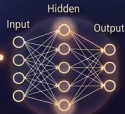


Clustering & K-Means

Logistic Regression



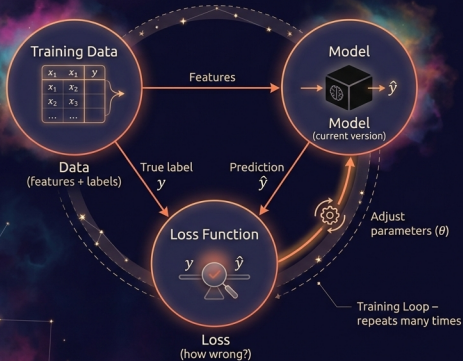
Logistic Regression



Neural Networks Preview

How a Model Learns

THE TRAINING CYCLE



INFERENCE (AFTER TRAINING)

After Training



New Input
(no label)

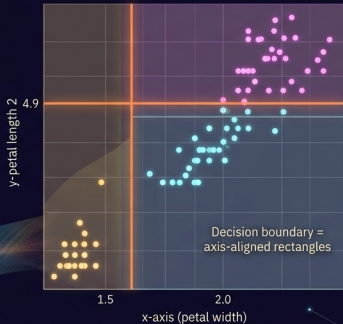
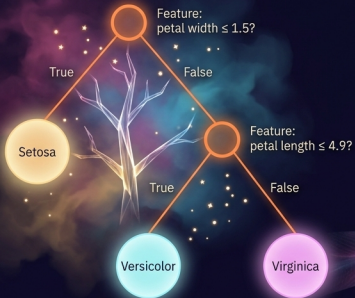


Trained Model
(locked)

e.g. Species: *versicolor*

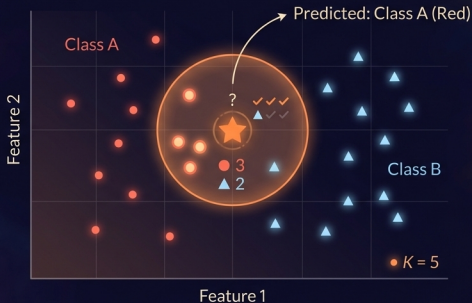
Prediction

Decision Tree

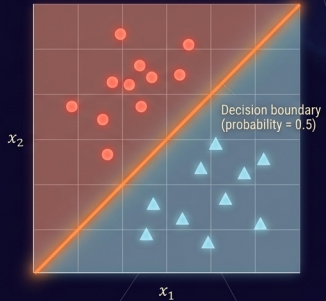
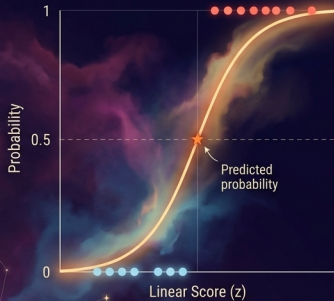


K-Nearest Neighbours (KNN)

Find the nearest neighbours, let them vote on a force field om query point.
Exactly $k = 5$ query point (the new example) - and indicatory majority wins.



Logistic Regression



Linear score $\rightarrow$ S-curve $\rightarrow$ Probability $\rightarrow$ Decision

Linear Regression

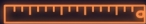


Loss Functions & Mean Squared Error

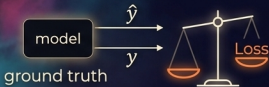
What is a Loss Function?

true value

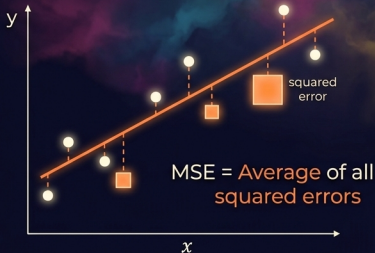
prediction



A loss function measures how wrong a prediction is.
The goal is to make it as small as possible.



Mean Squared Error (MSE) in action



MSE heavily punishes large errors – the model learns to avoid them.

Cross-Entropy Loss (Log Loss)

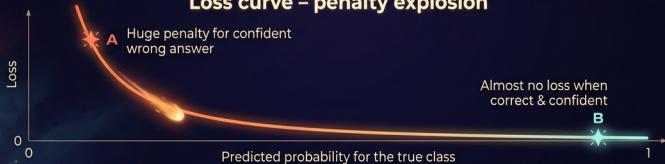
Good prediction (low loss)



Bad prediction (high loss)



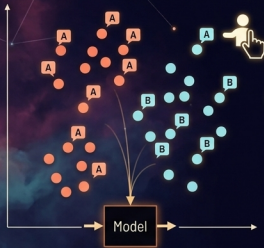
Loss curve – penalty explosion



A cosmic calmar-explose theorem using a slime "Neural Nebula" theme. Cross-entropy explodes when the model is confidently wrong. It forces the model to be both correct and well-calibrated.

Supervised vs Unsupervised Learning

Supervised



Training data has labels. The model learns from correct answers.

Both learn from data –
but only one has a teacher.

Both learn from data –
but only one has a teacher.

Unsupervised

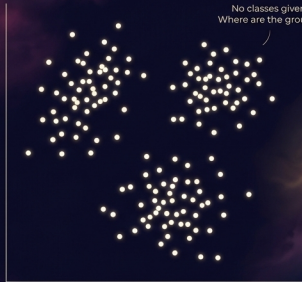


No labels. The algorithm finds
hidden patterns or groups.

What is Clustering?

Unlabelled Data

No classes given.
Where are the groups?

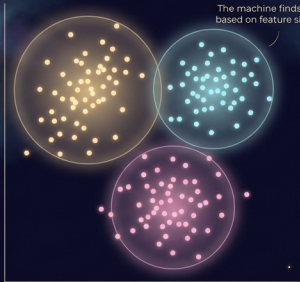


Find hidden
groups



Clusters Discovered

The machine finds groups
based on feature similarity.



Clustering = Grouping similar data points together, without any labels.

K-Means Clustering



Start



Assign



Move

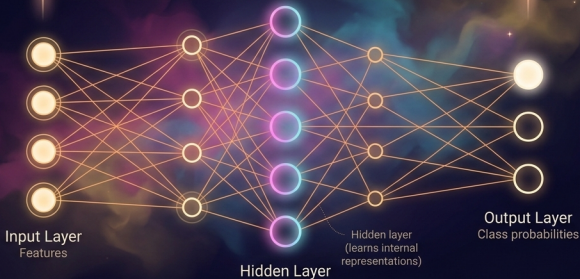


Repeat & Converge

K-Means alternates between assigning points and moving centers until the clusters stop changing.

Neural Networks: The Structure

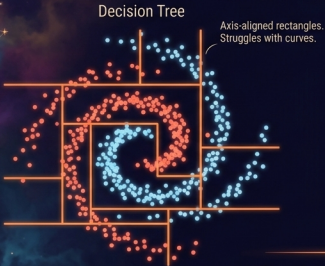
Stack many simple units → learn extremely complex patterns.



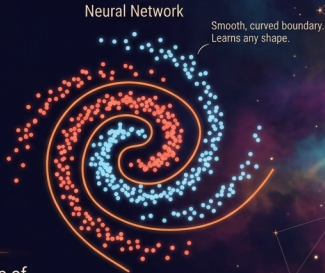
Next session: We'll train one on the Iris dataset.

Why Neural Networks? The Power

Decision Tree Boundary



Neural Network Boundary



When the world isn't made of
straight lines and rectangles...

Our Journey Through Machine Learning



You now speak the language of machine learning.